

Temporal changes in breast cancer incidence in South Asian women



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ABSTRACT

Background: Breast cancer in the UK resident population of South Asian ethnicity has been lower than that in indigenous women. Leicester has a large South Asian population and a breast cancer unit with comprehensive data on diagnosed cancers. This study analysed the annual incidence of new breast cancer diagnoses in females from 1998 to 2009 to determine any changes in recent years.

Methods: Ethnicity was known in over 98% of cases. Population denominators were based on published figures for 2001 and 2011, projected back to 1998. Age-adjusted directly standardised incidence rates were determined by ethnicity and broken down by invasive status and screening classification. Incidence rates were analysed using logistic regression in order to identify statistically significant effects of age, ethnicity, deprivation and year of diagnosis. Interactions with invasive status and screening classification were also investigated.

Results: At the start of the study period South Asian incidence was estimated to be 45% of that of the white population ($p < 0.001$); by the end of the period the difference was still significant ($p = 0.022$) but smaller, at 17%.

Conclusion: South Asians should no longer be considered at low risk of breast cancer.

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1. Introduction

People who originate from South Asia (India, Pakistan and Bangladesh) form the largest ethnic minority group in England and Wales: 3.9% at the time of the 2001 census increasing to 5.3% in 2011 [1]. In Leicester city, SA migrants and their descendants comprise more than one third of the population.

Breast cancer, in common with most cancers [2–4], has been found to be less frequent in SAs than in the UK white population, though most studies have suffered from high numbers of cases

with unknown ethnicity [5–8]. A Leicester study, which used cancer registration data from 1990–1999, and population data from the 1991 census, showed that SAs had much lower rates of breast cancer than the rest of the population, but there was a suggestion of increased risk in SAs with time [9]. More recently, a report of cancer incidence and survival by major ethnic group in England in 2002–2006 [5] showed an overall breast cancer age standardised incidence rate of between 122.4 and 125.7 (depending on how those with unknown ethnicity were analysed) per 100,000 for whites, compared to 59.7 to 92.3 for SAs.

Leicester's Breast Care Unit is centrally located in its catchment area, whose residents are infrequently treated elsewhere. High quality data has been collected prospectively on all breast cancers diagnosed since 1997, when breast cancer care was concentrated in specialist units. Regular data transfers to and from the cancer registry have helped maximise completeness of both data-sets [10]. Good information on patient ethnicity is also available.

Abbreviations: SA, South Asian; LLR, Leicester, Leicestershire and Rutland; UHL, University Hospitals of Leicester; EM QARC, East Midlands Quality Assurance Reference Centre; PCT, Primary Care Trust.

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2. Materials and methods

2.1. Cases

Patient data were provided by the Breast Care Unit of University Hospitals of Leicester (UHL), detailing diagnoses of invasive or ductal in situ (DCIS) breast cancer made between 1 January 1998 and 31 December 2009 in female residents of Leicester, Leicestershire and Rutland (LLR), as defined by local authority boundaries, which were stable over the study period. A small number of additional cases were supplied by East Midlands Quality Assurance Reference Centre (EM QARC) of screen-detected cancers in LLR residents diagnosed in neighbouring breast screening units. Other breast malignancies such as lymphoma, sarcoma or malignant phyllodes were excluded, as were contralateral breast cancers diagnosed during the time period, and local recurrence of cancer previously treated with breast conserving therapy. Data were linked to Primary Care Trust (PCT)-held patient records.

Self-reported ethnicity was obtained from routinely-collected hospital and PCT patient data. Where different ethnicities were reported on different occasions, but a clear majority coding was available, this was used. In occasional, difficult cases ethnicity was checked using name analysis, religion, diet, place of birth and names of relatives. Ethnicity was categorised as White, SA, Black and Other; the latter two groups were excluded from the analysis since the numbers were small. Ethnicity classification was achieved in over 98% of cases.

The data on breast cancer cases included postcode at diagnosis. This allowed the geographical area of residence to be identified; the Lower Super Output Area (LSOA), with boundaries that have been stable over the study period. National quintiles of the income domain of the English indices of deprivation (2004), defined at the LSOA level, were used to assign deprivation level, supported by other studies [4,8,11,12]. The LSOA is one of the smallest geographical units for which deprivation indices are routinely available and use of such small area geography reduces heterogeneity with regard to deprivation which may otherwise lead to inferences affected by 'ecological fallacy' [13].

2.2. Populations

To estimate and analyse incidence accurately, population estimates broken down by factors of interest are required; here age, year, ethnic group and deprivation quintile. These were derived from Office for National Statistics (ONS) estimates, based on data from the England and Wales 2001 and 2011 censuses [1].

The ONS population estimates used include mid-year estimates (by year, sex, LSOA and age), population estimates by ethnic group (by year, sex, local authority, ethnic group and age), and a specially commissioned 2001 census tabulation (C1238; by deprivation quintile, ethnic group, place of birth and age for LLR females) together with standard 2011 census tabulations. ONS population estimates by ethnic group are derived using a cohort component model [14]. This method projected estimates obtained from the 2001 census year-by-year adjusting for mortality, fertility, migration and ageing. Fertility and migration are based on ethnic group specific rates, whereas mortality rates are not. However, when the 2011 census data was published it emerged that considerably more SA population growth had occurred in some areas, including parts of LLR, than ONS had estimated. We compensated for this intercensal drift by adjusting the population estimates for the years 2002–2009 by a factor derived from the observed growth rates between 2001 and 2011 relative to the estimated growth over the study period. This adjustment was local authority, age, and ethnic group specific. These tabulations were combined and disaggregated using the technique of iterative

proportional fitting, or 'raking' [15,16] to generate population estimates for the years 2001 to 2009 broken down by deprivation quintile. The trends evident over those years were then projected back to provide consistent estimates for the years 1998 to 2000.

The majority of LLR South Asians, around 90%, are of Indian origin; numbers of Pakistani and Bangladeshi origin were too small to allow meaningful breakdown.

2.3. Missing data

In contrast with other studies where completeness of ethnicity data is low, fewer than 2% of cases were of unknown ethnicity. Nevertheless, appropriate treatment of missing data remains important since data are not, typically, missing at random, thus exclusion of cases with missing data could bias estimates. Furthermore, for accuracy, the cases observed (the numerator) and the population at risk (the denominator) must be consistent. The data on which ONS base their population estimates (the basis for the denominator) have missing values, notably ethnicity, imputed. A similar approach was therefore used with regard to the numerator, and missing values were imputed using multiple imputation by chained equations [17–19]. Analyses were performed in R [20], using the package mice [21]. 100 imputed datasets were generated using polytomous logistic regression.

2.4. Analysis

Directly standardised rates (DSRs) of incidence were calculated for each year, using the estimated white and SA populations, age-standardised according to the European standard population [22]. They are presented here as annual rates, in three-year bands, to smooth variations related to the three-year UK breast screening cycle. DSRs were also calculated broken down by invasive status (invasive versus in situ breast cancer) and screening classification. Screening classifications included screen-detected, interval (a cancer diagnosed between routine screening appointments), lapsed and non-attenders or other, which included cancers in patients who were outside the target screening age cohort: 50–64 (1998–2004) and 50–70 (2005 onwards).

Since differences in screening uptake by different ethnic groups could affect the apparent incidence of breast cancer, evidence for this was sought. Routinely-collected Breast Screening Program (BSP) data do not include ethnicity, so uptake of screening mammography in PCTs serving areas with widely different proportions of SA population was assessed.

Statistical modelling techniques were used to control for risk factors, estimate effects of interest and test for differences between groups over time. The Poisson generalised linear model is appropriate in this context [23], using an offset term to allow for the population at risk. In addition to considering age at diagnosis and ethnicity as factors, diagnosis year and deprivation quintile were included as covariates in analyses in order to establish annual trends and 'deprivation gradient', the latter representing the change in incidence among the least deprived relative to the most deprived [24]. Two-way interactions were also modelled. Incidence rate comparisons are reported as incidence rate ratios (IRRs).

Further analyses were performed to test whether modelled effects varied according to invasive status or screening classification (in those eligible for screening).

3. Results

From the period 1st January 1998 to 31st December 2009 a total of 8,318 patient records were supplied (7985 from UHL and 333 from EM QARC) of which 7,827 were found to meet the study

inclusion criteria. A summary of the socio-demographic and diagnostic characteristics of these patients is provided in Table 1.

Population modelling suggested that over the study period the female population of LLR grew from around 466,500 to 498,000; an increase of around 0.6% per annum. The proportion of this population classified as white fell from 87.1% to 80.3%, while the proportion classified as SA increased from 10.1% to 12.8%. This pattern was not uniform across all deprivation quintiles; there was a trend among SAs for greater population growth in lower deprivation areas, more so than the trend in whites. In the most deprived quintile the white population was estimated to fall from 68.4% to 53.9% and the SA population was estimated to increase from 24.9% to 29.4%. In the least deprived quintile the white population was estimated to fall from 95.4% to 91.6% while the SA population estimate increased from 3.1% to 4.9%.

Fig. 1 shows the percentage distribution of the estimated white and SA populations by age band, comparing 1998 with 2009. Both populations are seen to have aged over the twelve years, SAs more than whites. In 1998, the median age of white women, at around 40, was a decade greater than that of SAs; 23.0% of white, but only 9.4% of SA women, were aged over 60. By 2009 the gap in median age was smaller, though with white women still, on average, older: 26.2% of white and 12.5% of SA women were then aged over 60.

Uptake of screening mammography has been consistently high across LLR since the start of the NHS BSP. There was no difference in uptake in those from areas of high compared to low SA population over the later part of the study period (data broken down by geographical area was not available for the early years). Local data indicate that practices with more than 20% of SA population (range 21–75%) showed only a small increase in breast screening rates over two screening rounds. In 2008/2009, coverage data reported by the Information Centre showed that the performance in Leicestershire County and Rutland PCT was the highest in England at 84.3% [25]. The results should therefore be little affected by potential differences in screening uptake between groups.

Directly age-standardised breast cancer incidence rates for women of white and SA ethnicity are shown in Fig. 2. The estimated incidence among the SA population is seen to significantly increase from 64.4 per 100,000 in 1998–2000 to 110.6 per 100,000 in 2007–2009. The corresponding incidence rates among the white population were 111.7 and 128.2 per 100,000, respectively. The modelled DSR in SAs increased by 5.6% per annum; for white women the increase was 1.7% per annum.

Table 2 shows the significant two-way interactions found between diagnosis year and ethnicity ($p = 0.011$) and diagnosis year and age band ($p < 0.001$). The difference in incidence between the white and SA populations was highly significant at the start of the study ($p < 0.001$), with SA incidence estimated 45.2% lower than the incidence in the white population. At the end of the study period there still a significant difference in incidence by ethnicity ($p = 0.022$) but the differential had dropped to SA incidence being only 17.3% lower than for white women. There were significant annual increases in incidence among the SA population across most age bands except the youngest (15–39: $p = 0.315$; 40–49: $p = 0.012$; 50–59: $p = 0.003$; 60–69: $p < 0.001$; 70–79: $p = 0.003$; 80+: $p = 0.014$), whereas the only significant increase in incidence within the white population was in the age band 60–69 ($p < 0.001$).

Overall, incidence was not significantly associated with deprivation quintile ($p = 0.459$), however there were subtle age-related effects reflected in a significant age-deprivation interaction ($p = 0.003$). Table 2 shows a deprivation gradient, according to the statistical model identical for whites and SAs, which is significant for only the youngest and oldest age bands. For those aged 15–39 the incidence is significantly higher for the least, relative to the most, deprived quintile ($p = 0.004$). For those aged over 80, however, the gradient is reversed, with the most deprived having a significantly higher incidence ($p = 0.004$). For other age bands differences by deprivation are not statistically significant.

Table 1

Summary of study patient characteristics by ethnicity and by age band, year of diagnosis, deprivation quintile, place of birth or tumour invasive status. Percentage figures (in brackets) correspond to the proportion with each group classification.

Characteristic	Ethnicity					Total
	White	South Asian	Black	Other	Missing	
Total diagnoses	7120 (91%)	438 (6%)	48 (1%)	96 (1%)	125 (2%)	7827 (100%)
<i>By age</i>						
16–39	284 (83%)	36 (11%)	5 (1%)	10 (3%)	7 (2%)	342 (100%)
40–49	903 (85%)	97 (9%)	13 (1%)	23 (2%)	21 (2%)	1057 (100%)
50–59	2016 (91%)	126 (6%)	11 (<1%)	31 (1%)	35 (2%)	2219 (100%)
60–69	1738 (91%)	114 (6%)	11 (1%)	19 (1%)	26 (1%)	1908 (100%)
70–79	1266 (94%)	46 (3%)	6 (<1%)	9 (1%)	22 (2%)	1349 (100%)
80+	913 (96%)	19 (2%)	2 (<1%)	4 (<1%)	14 (1%)	952 (100%)
<i>By year of diagnosis</i>						
1998–2000	1567 (92%)	70 (4%)	7 (<1%)	19 (1%)	45 (3%)	1708 (100%)
2001–2003	1735 (92%)	85 (4%)	12 (1%)	24 (1%)	35 (2%)	1891 (100%)
2004–2006	1890 (91%)	118 (6%)	14 (1%)	25 (1%)	34 (2%)	2081 (100%)
2007–2009	1928 (90%)	165 (8%)	15 (1%)	28 (1%)	11 (1%)	2147 (100%)
<i>By deprivation quintile</i>						
1 (most)	656 (76%)	146 (17%)	27 (3%)	22 (3%)	11 (1%)	862 (100%)
2	863 (87%)	107 (11%)	5 (1%)	17 (2%)	5 (1%)	997 (100%)
3	1368 (92%)	76 (5%)	6 (<1%)	19 (1%)	15 (2%)	1484 (100%)
4	2179 (94%)	61 (3%)	6 (<1%)	23 (1%)	38 (2%)	2307 (100%)
5 (least)	2054 (94%)	48 (2%)	4 (<1%)	15 (1%)	56 (3%)	2177 (100%)
<i>By place of birth</i>						
Overseas	164 (34%)	250 (52%)	24 (5%)	39 (8%)	7 (1%)	484 (100%)
UK	3130 (97%)	14 (<1%)	4 (<1%)	12 (<1%)	63 (2%)	3223 (100%)
missing	3826 (93%)	174 (4%)	20 (<1%)	45 (1%)	55 (1%)	4120 (100%)
<i>By malignancy</i>						
DCIS	634 (90%)	52 (7%)	4 (1%)	9 (1%)	9 (1%)	708 (100%)
Invasive	6486 (91%)	386 (5%)	44 (1%)	87 (1%)	116 (2%)	7119 (100%)

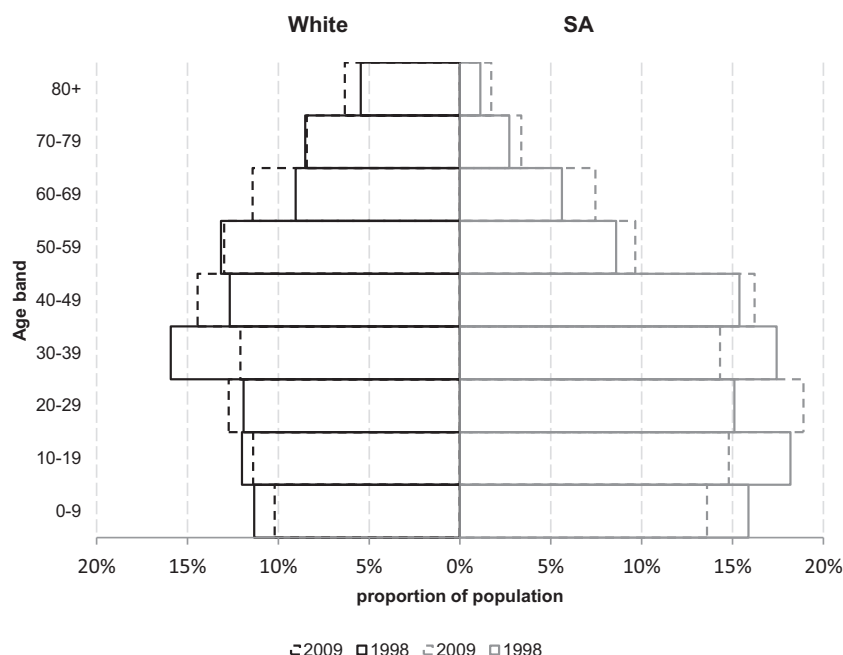


Fig. 1. Percentage of the white and SA estimated populations in each age band, for 1998 and 2009.

Fig. 3 shows the age-standardised incidence rates broken down by screening status: screen-detected, interval, and others. Significant interactions were found with both diagnosis year and age ($p < 0.001$) and diagnosis year and ethnicity ($p = 0.021$). Most of the increase in incidence in SA women relates to screen-detected or, to a lesser degree, interval cancers. Modelling incidence separately by screening classification confirms that in SAs there was a significant increase over the period studied, relative to that in white women, in both screen-detected ($p = 0.004$) and interval cancers ($p = 0.014$) but not in other categories ($p = 0.811$). Thus the bulk of the SA incidence increase occurred in women who were participating in the BSP.

Focusing on the age bands 50–59 and 60–69, within which screening typically occurs, there was a significant increase in incidence of screen-detected cancers among SAs within each age band (each $p < 0.001$), and also interval cancers among SA women aged 60–69 ($p = 0.003$) but not among those aged 50–59 ($p = 0.083$). The white population also showed significant, but smaller, increases in incidence of screen-detected cancers (each $p < 0.001$) but a significant reduction in interval cancers in the

50–59 age group ($p = 0.008$), with no change in incidence of interval cancers in the 60–69 age group ($p = 0.105$).

Considering tumour invasive status (invasive vs. DCIS; Fig. 4), the only significant findings were the expected interactions with age ($p = 0.013$), diagnosis year ($p = 0.005$) and screening classification ($p < 0.001$); most DCIS is diagnosed via the BSP (with an increased diagnosis rate over the years of the study), so is commoner in those in the screening age band. No significant interaction with ethnicity ($p = 0.097$) or deprivation ($p = 0.706$) was found.

4. Discussion

Correcting for confounding factors, an 82% increase in the overall incidence of breast cancer has been found among the SA population over the study period, which significantly exceeds the 21% increase in the white population. These results are robust, supported by reliable and near-complete data on cases and case ethnicity, and population data from two decennial census rounds. While ethnicity was known for 98.4% of cases overall, ascertainment improved from

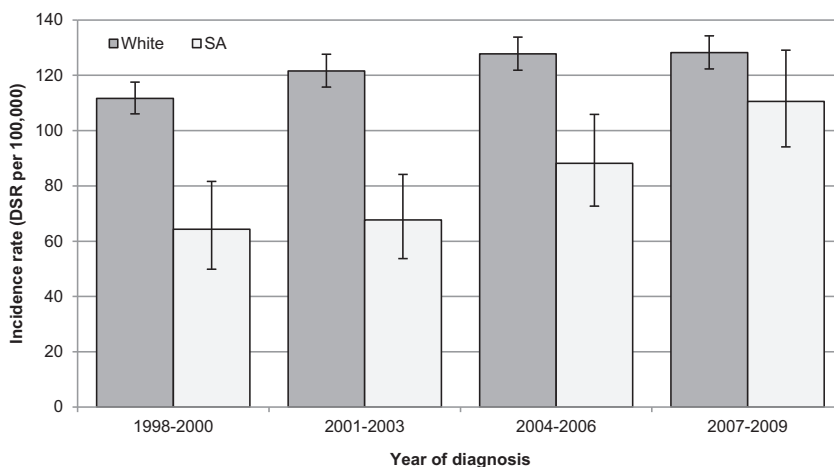


Fig. 2. Observed incidence of breast cancer (directly age-standardised rate per 100,000 European standard population) for females resident within LLR by ethnicity and year of diagnosis. Error bars indicate 95% confidence intervals.

Table 2

Modelled incidence rate ratios by age band with 95% confidence intervals (in brackets). The first column shows, for those of white ethnicity, the maximum likelihood estimate for each year's incidence relative to that of the previous year, to give the annual trend ($\pm$ CI) for the white population within each age band. The second column, similarly, shows the annual trends for the SA population. The third column presents the maximum likelihood estimates of the incidence rate of the least deprived relative to the most deprived, for the whole study period ($\pm$ CI), in each age band; these are the same for whites and SAs. Cells in bold indicate IRR values that vary significantly from 1; i.e. where the change in incidence is statistically significant at the 5% level.

Age band	Annual trend		Deprivation gradient
	White	South Asian	
15–39	0.984 (0.953–1.015)	1.021 (0.980–1.064)	1.810 (1.208–2.711)
40–49	1.003 (0.985–1.022)	1.042 (1.009–1.075)	1.069 (0.847–1.349)
50–59	1.008 (0.995–1.020)	1.046 (1.015–1.078)	1.050 (0.891–1.238)
60–69	1.050 (1.036–1.064)	1.090 (1.058–1.124)	1.049 (0.878–1.253)
70–79	1.012 (0.996–1.028)	1.050 (1.017–1.084)	1.101 (0.896–1.354)
80+	1.005 (0.987–1.024)	1.043 (1.009–1.079)	0.698 (0.546–0.891)

97.4% initially to 99.5%, which implies less certainty in incidence rates for the earlier years. This is reflected in the confidence intervals. Our initial analysis [26] had over-estimated the increase in breast cancer incidence in SAs because the ONS projections for this ethnic group proved, at the 2011 census, to have substantially under-estimated population growth, particularly for the SA population of LLR. We have allowed for this following publication of 2011 census results.

There was no significant difference over time, with ethnicity, in breast cancer diagnoses among symptomatic patients outside the screening age band, and lapsed and non-attenders; highly significant diagnosis year-ethnicity interactions were found for both screen-detected and interval cancers; the growth in incidence among the SA population has largely been with regard to those participating in the BSP. Since the uptake of screening mammography by the LLR population has been consistently good since the start of the BSP, and SAs constitute such a substantial minority, we

believe the results indicate a true increase in incidence, rather than an increase in screening attendance rate. It is, however, possible that a small part of the change could relate to some increase in screening uptake.

No overall difference in the incidence of DCIS was found in SAs compared to white women, contrasting with the Singapore screening program where a high proportion of DCIS was found [27]. Stage at diagnosis, treatment or outcomes have not yet been studied in any detail in the Leicester population, comparing SA with white breast cancer patients. Improved cancer-specific survival has been found in SAs in some studies, though not in others [28–31].

Apart from the youngest and oldest age bands, most of the LLR population, both white and SA, showed no effect of deprivation level on breast cancer incidence. This was unexpected, since breast cancer incidence has been shown to be higher in women of higher socio-economic groups [6,12,32], however these studies were unable to analyse jointly by ethnicity and deprivation, which may provide some explanation. There is also some evidence that any deprivation effect is reduced in communities that take part in breast screening [7]. Despite higher levels of deprivation in SA communities elsewhere in the UK, these findings may therefore apply to other UK SAs.

Multiple studies have shown marked variation in breast cancer incidence by race and ethnicity and most show that migration results in a change in incidence towards that of the indigenous population of the adopted residence. Some studies suggest that increased duration of residence changes risk progressively; some indicate that migration in childhood or early adulthood changes risk more than migration in later life; second generation migrants to high incidence countries may have a higher risk than their mothers' [33,34]. Since routine data on place of birth was scant, breast cancer incidence in immigrant SAs versus their descendants could not be compared in this study.

Such a rapid change in breast cancer incidence over a short, 12 year, time period is difficult to explain. In the Western world, high breast cancer incidence is attributed to population ageing, early detection, decreasing fertility, high prevalence of sedentary lifestyles and increasing post-menopausal obesity [35]. Total fertility of UK SAs has reduced, particularly for those of Pakistani and Bangladeshi origin [36]. The majority of LLR SAs are Indian in origin and for them the evidence is of a quite modest fertility reduction over the period 1987–2006. Women who are inactive,

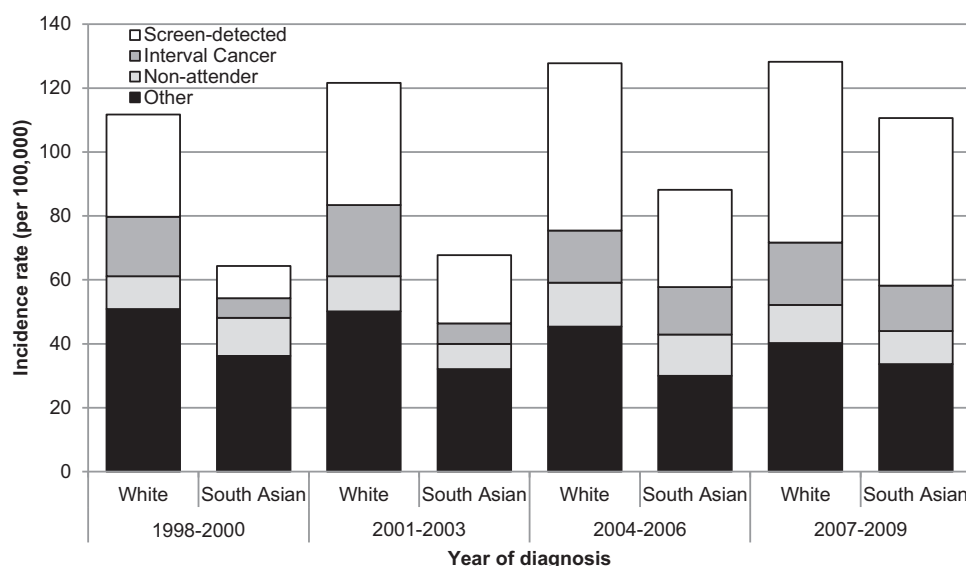


Fig. 3. Breast cancer incidence by ethnicity, year of diagnosis and screening status.

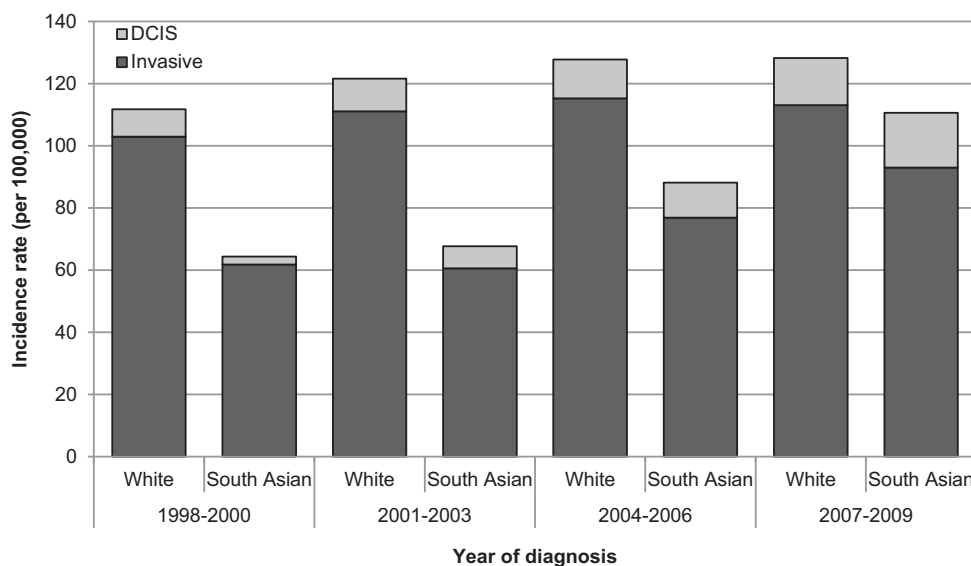


Fig. 4. Breast cancer incidence by ethnicity, year of diagnosis and tumour invasive status (invasive vs. DCIS).

obese or grossly obese have a substantially higher incidence, particularly of post-menopausal breast cancer, than those of lower mass [37–40]. Hormone replacement therapy (HRT) has been found to lead to a measurable increase in breast cancer risk over a few years, which then rapidly reduces with cessation of treatment [41]. Most factors are linked to hormonal influences, with about 90% of breast cancer incidence in the West being attributed to such hormone-level related factors [42]. We have incomplete data on HRT use, or body mass index, in LLR breast cancer patients, so have not attempted to take them into account in this analysis. Our impression is that HRT has been little used by SA women in LLR, though inactivity and obesity are recognised issues.

The LLR SA population has been subject to the effects of acclimatisation to a Western lifestyle, and the increased breast cancer detection rate evident in those educated in health issues and motivated to attend for screening mammography. The findings of this study have important implications for public health policy and the BSP within the UK, given its likely reproducibility outside of the immediate LLR population studied.

5. Conclusion

UK South Asians should no longer be considered at low risk of breast cancer.

Conflict of interest statement

The authors declare that there are no conflicts of interest.

Authorship contribution

We confirm that the revised manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us. All the listed authors have made substantial contributions to:

- Research design and/or acquisition, analysis or interpretation of data.
- Drafting the paper and revising it.
- Approving the submitted version.

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We understand that the corresponding author is the sole contact for the editorial process. She is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the corresponding author and which has been configured to accept email from anne.stotter@talktalk.net.

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